Data schema

English version of ../../database/schema.md.

Two databases (see architecture/system-context.md):

Database	Purpose	Where it is described
PostgreSQL	domain data and analytics aggregates	db/schema.sql, backend/migrations/*.sql, backend/app/schema.py
SQLite	accounts, sessions, keys	backend/auth_schema.sql (db/schema.sql is copied into the image as the auth schema)

Data ownership

Domain	Writer	Reader
Source snapshot (accounts, services, transactions)	snapshot import and AEZA_SNAPSHOT_PATH	analytics domain
Aggregates and risk metrics	analytics domain (backend/app/domain/*)	API, export, AI tools
AI insights	the AI service (after tool validation)	API, reports
Actions and scenarios	domain/actions.py, domain/scenarios through the API	API, AI tools
Users and sessions	backend/auth (SQLite)	API

Rules

1. **Money is integer minor units plus a currency.** Arithmetic lives in backend/app/domain/money.py; float is forbidden in revenue calculations.
2. **Provenance is mandatory.** Every aggregate knows which snapshot it came from (domain/provenance.py), otherwise a number cannot be explained.
3. **The risk score is explainable:** the contribution of every factor is stored, not just the total.
4. **The snapshot is replaced as a whole** (PUT /api/v1/source/snapshot): it is a re-indexing operation, not a partial UPDATE of arbitrary rows.

Verification

- Applying the schema to an empty database: make compose-up → the API start runs schema.ensure_schema() (column reconciliation plus the migration registry).
- Repeated start: the schema_migrations registry prevents a second application.
- Data: python backend/scripts/load_snapshot.py and backend/tests/test_api.py.